TRIBHUVAN UNIVERSITY

INSTITUTE OF ENGINEERING

THAPATHALI CAMPUS

A LAB REPORT
ON
GENERATION OF DSB-SC, DSB-FC and SSB-SC SIGNALS

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TITLE: GENERATION OF DSB-SC, DSB-FC and SSB SIGNALS

OBJECTIVES

- To understand the analog amplitude modulation.
- To perform DSB-FC, DSB-SC and SSB-SC amplitude modulation on baseband signals.

TOOLS REQUIRED

- Python 3.12
- Numpy
- Matplotlib
- Scipy
- Jupyter Notebook

THEORY

Signal Modulation

Signal Modulation refers to the process of encoding the information/data (baseband signal) onto a carrier frequency (a high frequency signal) for transmission over a long distance.

The encoding process alters any of the properties of the carrier signal (i.e. amplitude, frequency, or phase) according to the baseband signal that imprints the information onto the carrier signal.

The modulation of baseband signal into high frequency carrier signal provides several benefits:

- Provides better transmission range by shifting the signal to high frequency bands.
- Reduces power loss during propagation and allows efficient radiation using practical antennas.
- Allows multiple signals to share the same medium using different carrier frequencies (Frequency Division Multiplexing).
- Improves signal-to-noise ratio (SNR) over long distances.

Types of Modulation

Major types of modulations are:

1. **Analog Modulation**

   Analog modulation is a method in which a continuous carrier signal is changed according to a continuous message signal. The message signal varies smoothly with time, such as speech or music. In analog modulation, one property of the carrier wave amplitude, frequency, or phase is varied continuously to represent the information. Common examples include AM and FM radio broadcasting.

   a. **Amplitude Modulation** It is the process of varying the amplitude of the carrier wave according to the message signal (baseband signal) to modulate the signal.

   b. **Frequency Modulation** It is the process of varying the frequency of the carrier wave according to the message signal (baseband signal) to modulate the signal.

   c. **Phase Modulation** It is the process of varying the phase of the carrier wave according to the message signal (baseband signal) to modulate the signal.

2. **Digital Modulation**

   Digital modulation is a method in which a carrier signal is changed according to digital data, typically in the form of binary values 0 and 1. Instead of continuous variation, the carrier's properties change in discrete steps. This type of modulation is widely used in modern communication systems such as mobile communication, Wi-Fi, and digital television.

   a. **ASK (Amplitude Shift Keying)** It is the process of changing the amplitude of the carrier wave according to the digital data '0' or '1' where the carrier is present when data is '1' and absent when data is '0'.

   b. **FSK (Frequency Shift Keying)** It is the process of varying the amplitude of carrier waves according to digital data '0' or '1'.

c. **PSK (Phase Shift Keying)** It is the process of varying the phase of the carrier wave.

   i. For binary data (1 bit symbol) – Binary PSK (0 and 180 degrees)
   ii. For 2-bit symbols – Quadrature PSK (4 phase shifts)

d. **QAM (Quadrature Amplitude Modulation)** It is the process that combines both variations in amplitude and phase shift to send multiple bits per symbol increasing the data rate.

Hilbert Transform

Hilbert Transform is a mathematical operation used in signal processing that provides a phase shift of +90° to all the negative frequencies and a phase shift of −90° to all the positive frequencies of the signal. It creates a signal that is orthogonal to the original signal.

Amplitude Modulation

Amplitude modulation refers to the process of changing the amplitude of the carrier signal with respect to the baseband or message signal.

Let the carrier signal be:

$$C(t) = A_c \cos(2\pi f_c t)$$

Modulating signal:

$$m(t) = A_m \sin(2\pi f_m t)$$

There are various types of Amplitude Modulation:

DSB-FC (Double Side Band – Full Carrier)

When a signal is amplitude-modulated, the signal is split into two bands i.e. upper band and lower band. If both the upper band and lower band signals are sent along with the carrier signal, it is known as the DSB-FC amplitude modulation.

The basic expression for DSB-FC modulated signal is:

$$C_m(t) = A_c\,(1 + k_a m(t)) \cos(\omega_c t)$$

$$C_m(t) = A_c\,(1 + k_a A_m \sin(\omega_m t))\cos(\omega_c t)$$

For proper DSB-FC modulation, the value of $\mu = k_a A_m$ must be less than 1 to preserve the envelope of the message signal.

The DSB-FC modulated signal can be demodulated asynchronously with envelop detection technique.

Key Characteristics:

- In DSB-FC, the amplitude of the carrier is controlled by the message signal.
- The information is encoded in the amplitude of the carrier, not in its frequency.
- The carrier itself does not carry information, but it helps in simple demodulation, such as envelope detection.
- In real systems, AM transmitters are designed with a strong carrier, which ensures that envelope detection works reliably.
- The message amplitude is usually limited to prevent distortion and spectral spreading.
- The carrier frequency is chosen high enough to allow efficient transmission over long distances and to satisfy antenna size constraints.
- The modulation index is typically kept less than 1 to ensure a clean signal and avoid nonlinear distortion.

DSB-SC (Double Side Band – Suppressed Carrier)

If both the upper sideband and lower sideband signals are transmitted but the carrier signal is suppressed, the modulation is called Double Sideband Suppressed Carrier (DSB-SC) amplitude modulation. In this scheme, the carrier does not carry any information, and all the transmitted power is contained in the sidebands.

The basic expression for a DSB-SC modulated signal is:

$$s(t) = m(t) \cdot c(t)$$

where $m(t)$ is the message signal and $c(t)$ is the carrier signal. For a single-tone message:

$$m(t) = A_m \sin(2\pi f_m t), \quad c(t) = A_c \cos(2\pi f_c t)$$

Hence, the DSB-SC modulated signal can be expressed as:

$$s(t) = A_m A_c \sin(2\pi f_m t) \cos(2\pi f_c t)$$

Key Characteristics:

- The carrier is suppressed and not transmitted.
- Both upper and lower sidebands carry the message information.
- Envelope detection cannot be used; coherent detection is required.
- Power is more efficiently used compared to DSB-FC since no carrier power is wasted.
- The total bandwidth of the DSB-SC signal is $2f_m$.

Single Sideband - Suppressed Carrier (SSB-SC)

If only one of the sidebands (either upper or lower) is transmitted while suppressing the carrier and the other sideband, the modulation scheme is known as Single Sideband Suppressed Carrier (SSB-SC) amplitude modulation. In this method, the transmitted signal occupies minimum bandwidth and utilizes power more efficiently compared to DSB-FC and DSB-SC.

In SSB-SC modulation, the carrier does not carry information and is not transmitted. All the information is contained in a single sideband, either the Upper Sideband (USB) or the Lower Sideband (LSB). As a result, SSB-SC requires coherent detection at the receiver.

Methods of SSB-SC Generation

The commonly used methods for generating SSB-SC signals are:

Filter Method (Frequency Discrimination Method): In this method, a DSB-SC signal is first generated. A highly selective bandpass filter is then used to remove one of the sidebands while allowing the other sideband to pass. The output of the filter is an SSB-SC signal. This method requires sharp filters, especially when the message frequency is low.

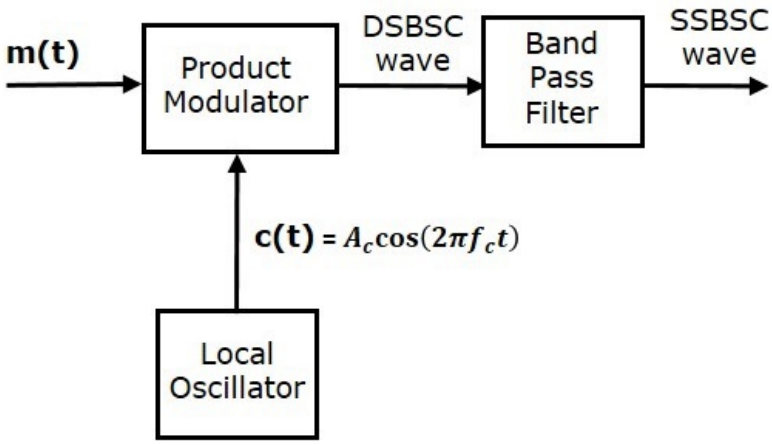

Figure 1: SSB-SC generation using Frequency (Filter) Discrimination Method

Phase Discrimination Method: In this method, phase shifters are used to introduce a 90° phase shift to the message signal and the carrier signal. The resulting signals are combined in such a way that one sideband cancels out due to phase opposition, leaving only the desired sideband. This method avoids the need for sharp filters and is widely used in practical SSB systems.

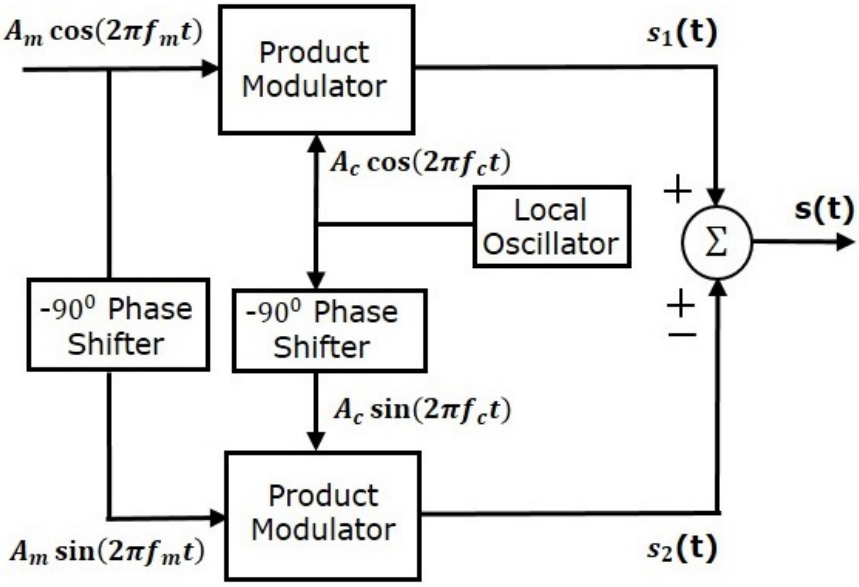

Figure 2: SSB-SC generation using Phase Discrimination Method

Mathematical Representation

For a single-tone message signal $m(t) = A_m \sin(2\pi f_m t)$ and a carrier signal $c(t) = A_c \cos(2\pi f_c t)$, the SSB-SC signal can be expressed as:

Upper Sideband (USB)

$$s_{\text{USB}}(t) = \frac{A_m A_c}{2} \cos\left[2\pi(f_c + f_m)t\right]$$

Lower Sideband (LSB)

$$s_{\text{LSB}}(t) = \frac{A_m A_c}{2} \cos\left[2\pi(f_c - f_m)t\right]$$

Key Characteristics:

- Only one sideband is transmitted; the carrier and the other sideband are suppressed.
- Bandwidth required is f_m, which is half that of DSB-FC and DSB-SC.
- Highly power efficient since no power is wasted on carrier and redundant sideband.
- Requires coherent detection at the receiver.
- Widely used in long-distance and bandwidth-constrained communication systems.

PROCEDURE

1. Define the sampling parameters.
2. Generate time instances of sampling.
3. Create a sinc signal.
4. Define a cosine wave carrier signal of high frequency (must satisfy Nyquist Sampling Theorem for defined sampling frequency).
5. Perform Modulation.
   (a) DSB-FC
   (b) DSB-SC
   (c) SSB-SC
6. Plot the modulated signals with respect to time.
7. Plot the power spectrum of the modulated signals.
8. Compare and analyse the outputs of all modulated signals.
9. Try the phase discrimination method for SSB signal generation.

IMPLEMENTATION

Double Side Band – Full Carrier (DSB-FC)

The DSB-FC signal is implemented by varying the amplitude of a high-frequency carrier in proportion to the message signal. In the implementation, a low-frequency sinusoidal message signal is generated and added to a constant carrier term to ensure that the carrier is transmitted in full. The resulting signal contains the carrier component along with both upper and lower sidebands.

1. **Sampling frequency (Fs)**
   Determines how finely the signal is sampled. Must satisfy Nyquist: $Fs > 2\,f_m$. Effect: higher Fs gives smoother signal and more accurate frequency spectrum.

2. **Duration (T)**
   Total time of the signal. Effect: longer T improves frequency resolution in FFT.

3. **Message amplitude (Am)**
   Amplitude of the baseband signal. Effect: larger Am increases envelope height; if too large, over-modulation occurs.

4. **Carrier amplitude (Ac)**
   Amplitude of the high-frequency carrier. Effect: higher Ac produces stronger signal and ensures envelope detection works. Typically, $Ac > Am$.

5. **Message frequency (fm)**
   Frequency of the baseband signal. Effect: higher fm $\rightarrow$ faster envelope changes; bandwidth = $2f_m$.

6. **Carrier frequency (fc)**
   Frequency of the carrier, must be much higher than fm. Effect: shifts signal in frequency domain; envelope shape unchanged.

7. **Modulation index / amplitude sensitivity (ka)**
   Scales the message relative to the carrier. Effect: higher ka strengthens modulation; ka > 1 can cause over-modulation.

Code:

```python
import numpy as np
import matplotlib.pyplot as plt

# Defining Sampling Parameters
Fs = 10000                  # Sampling frequency (Hz)
T = 0.01                    # Duration of signal (seconds)
t = np.arange(0, T, 1/Fs)   # Time vector (sampling instances)
ka = 1                      # amplitude sensitivity

# Fs must satisfy Nyquist: Fs > 2*fm

# Baseband Signal (Message Signal)
# Single-tone baseband signal m(t) = Am*cos(2*pi*fm*t)
Am = 1                      # Amplitude of message signal
fm = 50                     # Frequency of message signal (Hz)
m_t = Am * np.sin(2 * np.pi * fm * t)

# Carrier Signal (High-frequency carrier)
Ac = 5                      # Carrier amplitude
fc = 1000                   # Carrier frequency (Hz)
c_t = Ac * np.cos(2 * np.pi * fc * t)

# DSB-FC Modulation (The carrier is transmitted in full)
# Formula: s(t) = Ac * [1 + ka * m(t)] * cos(2*pi*fc*t)
s_dsbfc = Ac * (1 + ka * m_t) * np.cos(2 * np.pi * fc * t)

# Ploting Time-Domain Signals
plt.figure(figsize=(12, 8))

plt.subplot(3,1,1)
plt.plot(t, m_t, 'b', linewidth=1.5)
plt.title('Message Signal m(t)')
plt.xlabel('Time [s]')
```

```python
plt.ylabel('Amplitude')
plt.grid(True)

plt.subplot(3,1,2)
plt.plot(t, c_t, 'r', linewidth=1.5)
plt.title('Carrier Signal c(t)')
plt.xlabel('Time [s]')
plt.ylabel('Amplitude')
plt.grid(True)

plt.subplot(3,1,3)
plt.plot(t, s_dsbfc, 'g', linewidth=1.5)
plt.title('DSB-FC Modulated Signal s(t)')
plt.xlabel('Time [s]')
plt.ylabel('Amplitude')
plt.grid(True)

plt.tight_layout()
plt.show()

# Frequency-Domain Analysis
N = len(s_dsbfc)
S_f = np.fft.fft(s_dsbfc)/N        # FFT and normalization
f = np.fft.fftfreq(N, 1/Fs)        # Frequency axis

f_pos = f[:N//2]                   # Take only positive frequencies
S_f_pos = 2 * np.abs(S_f[:N//2])   # Multiply by 2 to account for symmetry

plt.figure(figsize=(10,4))
plt.plot(f_pos, S_f_pos, 'm', linewidth=1.5)
plt.title("Magnitude Spectrum of DSB-FC Signal")
plt.xlabel("Frequency [Hz]")
plt.ylabel("Magnitude")
plt.grid(True)
plt.show()
```

Output:

The above plots are obtained using the following parameters: sampling frequency $Fs = 10$ kHz, signal duration $T = 0.01$s, message amplitude $A_m = 1$, carrier amplitude $A_c = 5$, message frequency $f_m = 50$ Hz, carrier frequency $f_c = 1000$ Hz, and modulation index $k_a = 1$.

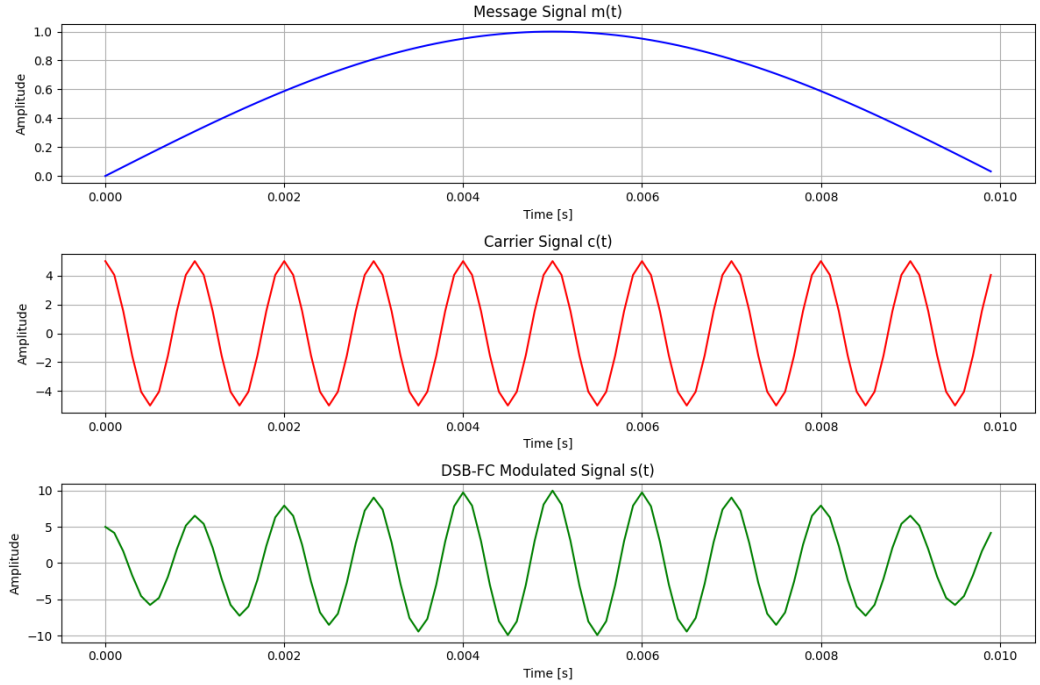

Figure 3: Plots of (a) Message Signal, (b) Carrier Signal, and (c) DSB-FC Modulated Signal

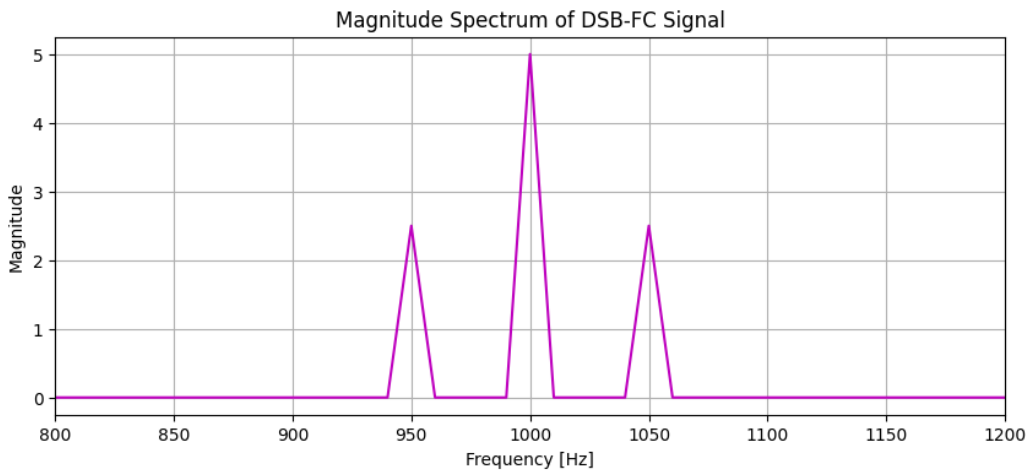

Figure 4: Plots of DSB-FC spectrum

Observations:

From the time-domain plot (Figure 3):

- The message signal $m(t)$ is a low-frequency sinusoid.
- The carrier signal $c(t)$ is a high-frequency cosine wave.
- The DSB-FC modulated signal $s(t)$ shows that the amplitude of the carrier varies according to the message signal, forming an envelope that matches $m(t)$.

From the frequency-domain plot (Figure 4):

- The spectrum shows a peak at the carrier frequency $f_c = 1000$ Hz.
- Two sidebands appear at $f_c \pm f_m$ (i.e., 950 Hz and 1050 Hz), carrying the message information.
- The bandwidth of the DSB-FC signal is $2f_m = 100$ Hz.

Variations of Different Parameters:

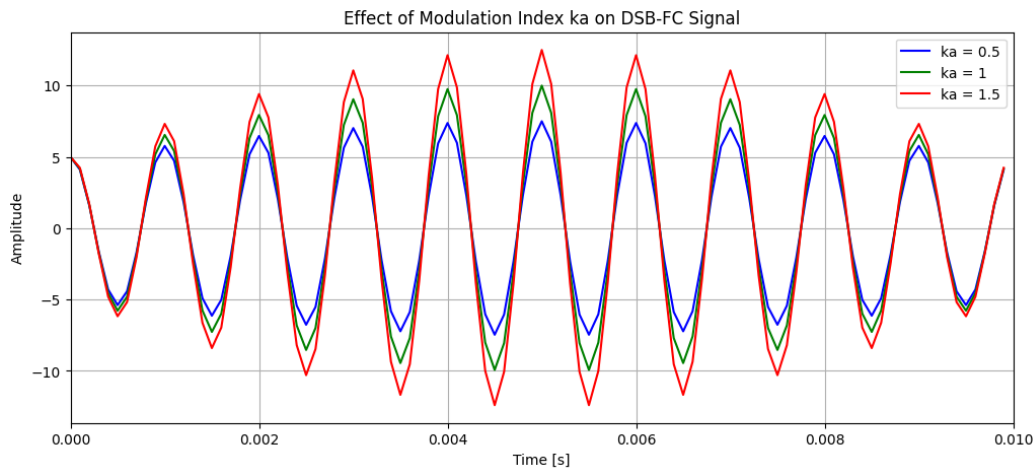

Figure 5: Effect of modulation index k_a on DSB-FC signal

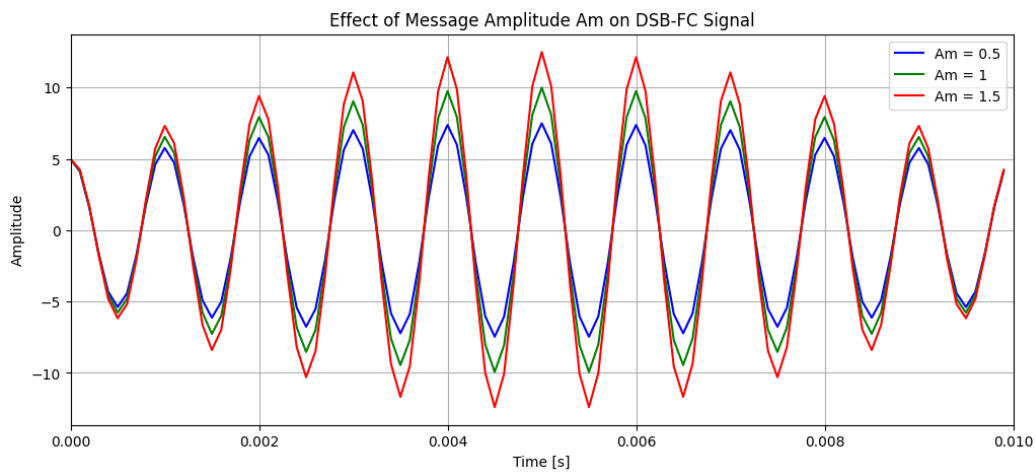

Figure 6: Effect of message amplitude A_m on DSB-FC signal

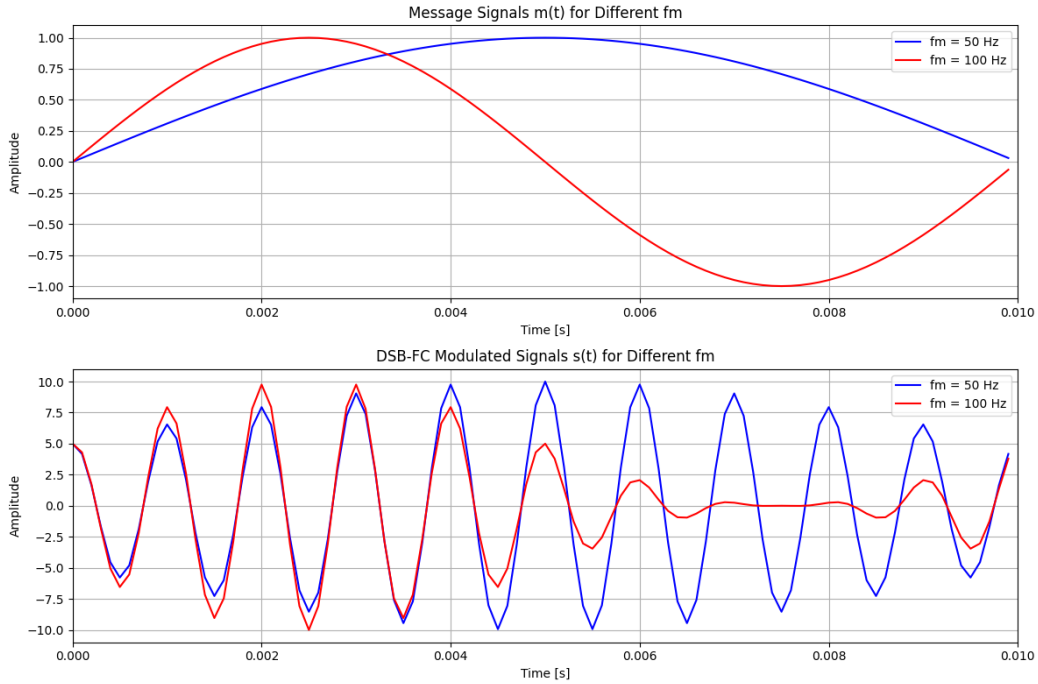

Figure 7: Effect of message frequency f_m on DSB-FC signal in time domain

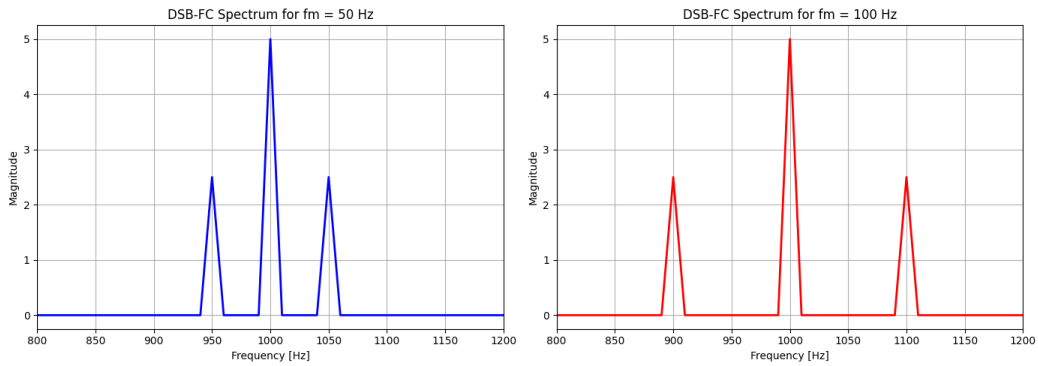

Figure 8: Effect of message frequency f_m on DSB-FC spectrum

- DSB-FC spectrum shows a strong carrier peak at f_c and two smaller sidebands at $f_c \pm f_m$.
- fm = 50 Hz: envelope oscillates slowly; sidebands at 950 Hz and 1050 Hz, bandwidth = 100 Hz.
- fm = 100 Hz: envelope oscillates faster; sidebands at 900 Hz and 1100 Hz, bandwidth = 200 Hz.
- Higher message frequency increases both the envelope oscillation rate and the spectrum bandwidth.

Double Side Band – Suppressed Carrier (DSB-SC)

The DSB-SC signal is implemented using a balanced modulation technique, where the message signal is directly multiplied with the carrier signal. This multiplication suppresses the carrier component while producing both upper and lower sidebands. In the implementation, time-domain plots are used to visualize the modulated waveform, and frequency-domain analysis is carried out to verify the absence of the carrier and the presence of symmetric sidebands around the carrier frequency.

Code:

```
import numpy as np
import matplotlib.pyplot as plt

# Sampling Parameters
Fs = 10000                    # Sampling frequency (Hz)
T = 0.1                      # Duration of signal (seconds)
t = np.arange(0, T, 1/Fs)   # Time vector

# Baseband signal (message)
Am = 1                        # Message amplitude
fm = 50                       # Message frequency (Hz)
m_t = Am * np.sin(2 * np.pi * fm * t)

# Carrier signal
fc = 1000                            # Carrier frequency (Hz)
c_t = np.cos(2 * np.pi * fc * t)   # amplitude = 1 for DSB-SC

# DSB-SC Modulated signal
s_dsbsc = m_t * c_t   # No carrier term added

# Time-domain plots
plt.figure(figsize=(12, 8))

plt.subplot(3,1,1)
plt.plot(t, m_t, 'b', linewidth=1.5)
```

```
plt.title('Message Signal m(t)')
plt.xlabel('Time [s]')
plt.ylabel('Amplitude')
plt.xlim(0, 0.05)
plt.grid(True)

plt.subplot(3,1,2)
plt.plot(t, c_t, 'r', linewidth=1.5)
plt.title('Carrier Signal c(t)')
plt.xlabel('Time [s]')
plt.ylabel('Amplitude')
plt.xlim(0, 0.05)
plt.grid(True)

plt.subplot(3,1,3)
plt.plot(t, s_dsbsc, 'g', linewidth=1.5)
plt.title('DSB-SC Modulated Signal s(t)')
plt.xlabel('Time [s]')
plt.ylabel('Amplitude')
plt.xlim(0, 0.05)
plt.grid(True)

plt.tight_layout()
plt.show()

# Frequency-domain analysis
N = len(s_dsbsc)
S_f = np.fft.fft(s_dsbsc)/N
f = np.fft.fftfreq(N, 1/Fs)

f_pos = f[:N//2]
S_f_pos = 2 * np.abs(S_f[:N//2])

plt.figure(figsize=(10,4))
plt.plot(f_pos, S_f_pos, 'm', linewidth=1.5)
```

```
plt.title("Magnitude Spectrum of DSB-SC Signal")
plt.xlabel("Frequency [Hz]")
plt.ylabel("Magnitude")
plt.grid(True)
plt.xlim(fc - 200, fc + 200)
plt.show()
```

Output:

The above plots are obtained using the following parameters: sampling frequency $F_s = 10$ kHz, signal duration $T = 0.1$ s, message amplitude $A_m = 1$, carrier amplitude $A_c = 1$, message frequency $f_m = 50$ Hz, and carrier frequency $f_c = 1000$ Hz.

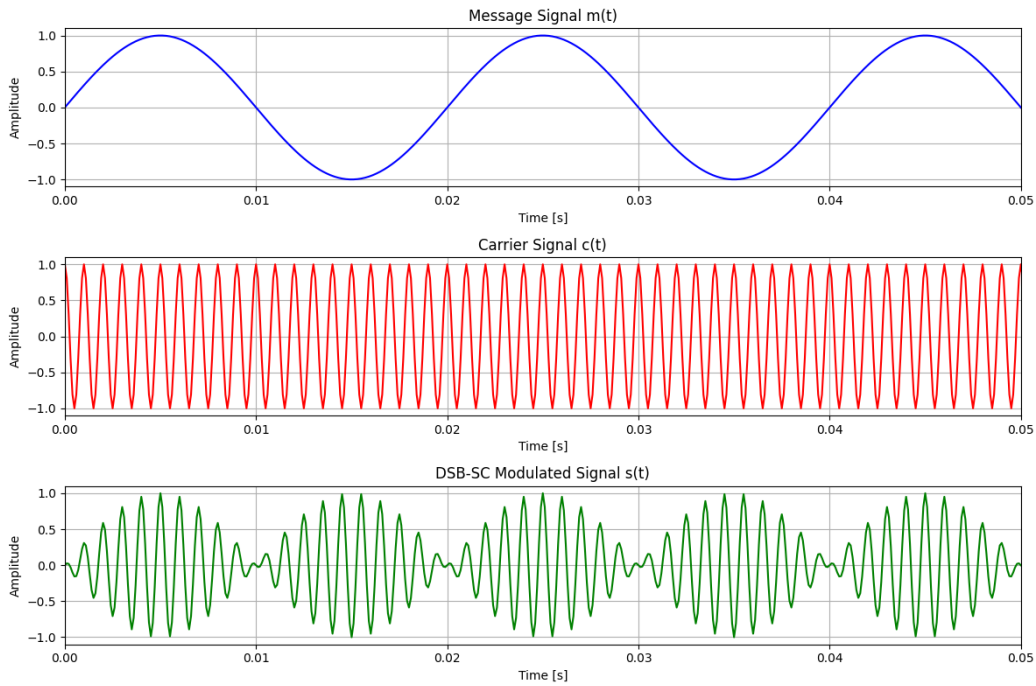

Figure 9: Plots of (a) Message Signal, (b) Carrier Signal, and (c) DSB-SC Modulated Signal

Observations:

Although DSB-SC modulation theoretically suppresses the carrier completely, a small spectral component near the carrier frequency is observed in the FFT when the signal duration is reduced to $T = 0.01$ s. This occurs due to limited frequency resolution and spectral leakage associated with finite-duration signals. Increasing the signal duration improves the frequency resolution of the FFT, thereby reducing this effect and clearly revealing only the upper and lower sidebands.

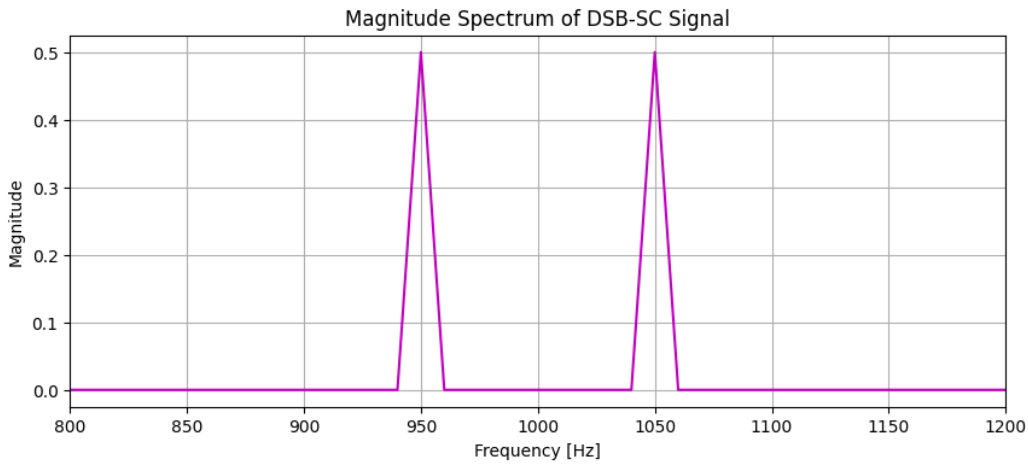

Figure 10: Plots of DSB-SC spectrum (0.1s)

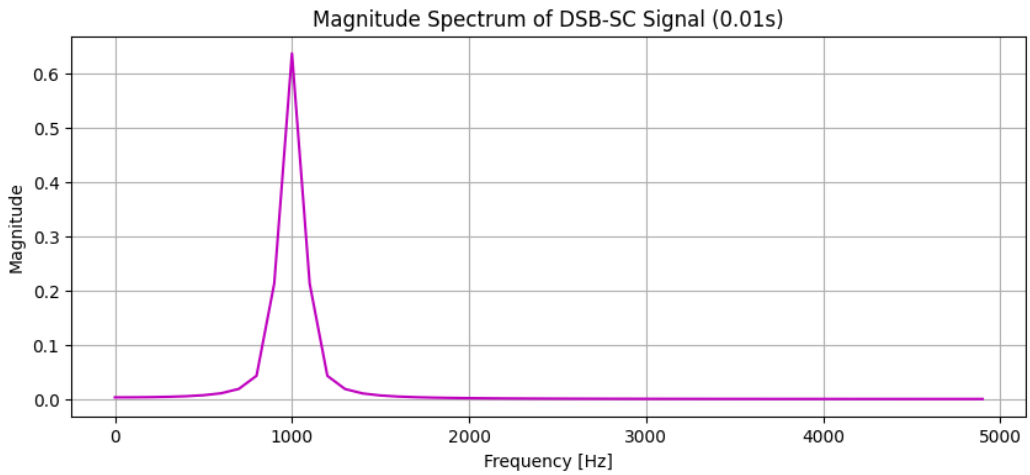

Figure 11: Plots of DSB-SC spectrum (0.01s)

Single Side Band – Suppressed Carrier (SSB-SC)

The SSB-SC signal is implemented by first generating a DSB-SC signal and then eliminating one of the sidebands using frequency-domain processing. In the implementation, the message signal is modulated with the carrier to produce a DSB-SC signal, after which either the upper or lower sideband is retained while the other is suppressed. Time-domain plots are used to observe the modulated waveform, and frequency-domain analysis is performed to confirm the presence of only a single sideband with the carrier fully suppressed.

Code (Filter Method):

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.signal import butter, filtfilt
```

```python

# Sampling parameters
Fs = 10000                # Sampling frequency (Hz)
T = 0.1                   # Duration of signal (seconds)
t = np.arange(0, T, 1/Fs)

# Message signal
Am = 1
fm = 50
m_t = Am * np.sin(2 * np.pi * fm * t)

# Carrier signal
Ac = 1
fc = 1000
c_t = Ac * np.cos(2 * np.pi * fc * t)

# Generate DSB-SC signal
s_dsbsc = m_t * c_t  # Suppressed carrier

# Design band-pass filter to select upper sideband (USB)
lowcut = fc            # Hz
highcut = fc + 2*fm    # Hz

# Butterworth bandpass filter
order = 6
b, a = butter(order, [lowcut/(Fs/2), highcut/(Fs/2)], btype='band')
s_usb = filtfilt(b, a, s_dsbsc)

# Plot time-domain signals
plt.figure(figsize=(12,8))

plt.subplot(3,1,1)
plt.plot(t, m_t, 'b', linewidth=1.5)
plt.title('Message Signal m(t)')
plt.xlabel('Time [s]')
```

```
plt.ylabel('Amplitude')
plt.grid(True)

plt.subplot(3,1,2)
plt.plot(t, s_dsbsc, 'g', linewidth=1.5)
plt.title('DSB-SC Signal s(t)')
plt.xlabel('Time [s]')
plt.ylabel('Amplitude')
plt.grid(True)

plt.subplot(3,1,3)
plt.plot(t, s_usb, 'r', linewidth=1.5)
plt.title('SSB-SC Signal (Upper Sideband)')
plt.xlabel('Time [s]')
plt.ylabel('Amplitude')
plt.grid(True)

plt.tight_layout()
plt.show()

# Frequency-domain analysis
N = len(s_usb)
S_f = np.fft.fft(s_usb)/N
f = np.fft.fftfreq(N, 1/Fs)
f_pos = f[:N//2]
S_f_pos = 2 * np.abs(S_f[:N//2])

plt.figure(figsize=(10,4))
plt.plot(f_pos, S_f_pos, 'm', linewidth=1.5)
plt.title("Magnitude Spectrum of SSB-SC Signal (USB)")
plt.xlabel("Frequency [Hz]")
plt.ylabel("Magnitude")
plt.grid(True)
plt.xlim(fc-50, fc+2*fm+50)
plt.show()
```

Output:

The above plots are obtained using the following parameters: sampling frequency F_s = 10 kHz, signal duration $T = 0.1$ s, message amplitude $A_m = 1$, carrier amplitude $A_c = 1$, message frequency $f_m = 50$ Hz, and carrier frequency $f_c = 1000$ Hz. Butterworth band-pass filter order = 6.

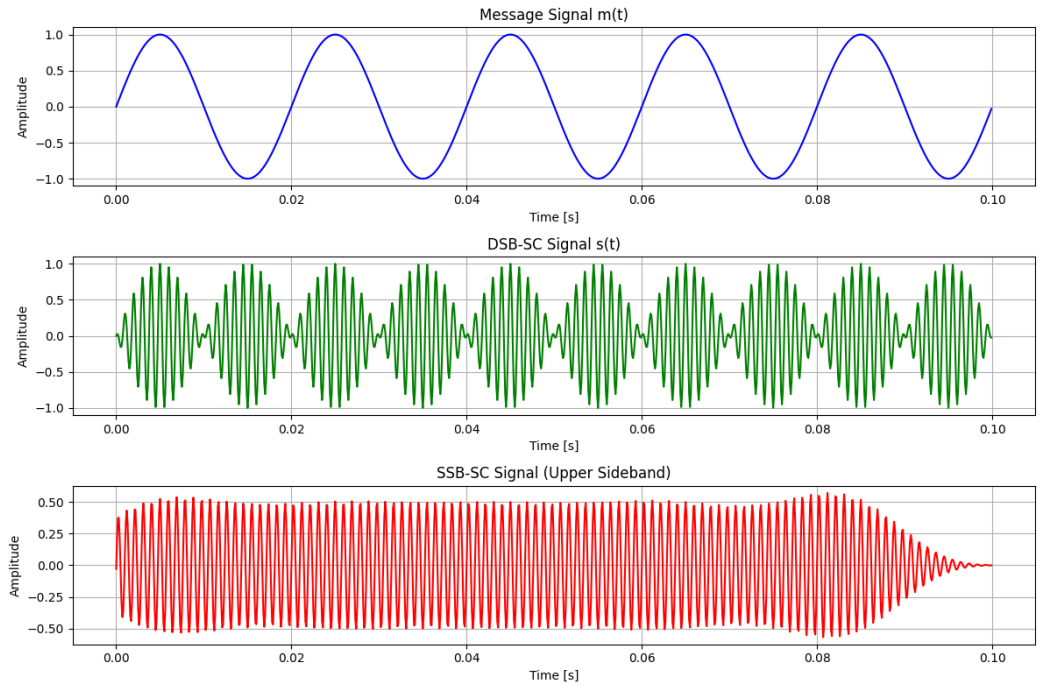

Figure 12: Plots of (a) Message Signal, (b) Carrier Signal, and (c) SSB-SC Modulated Signal

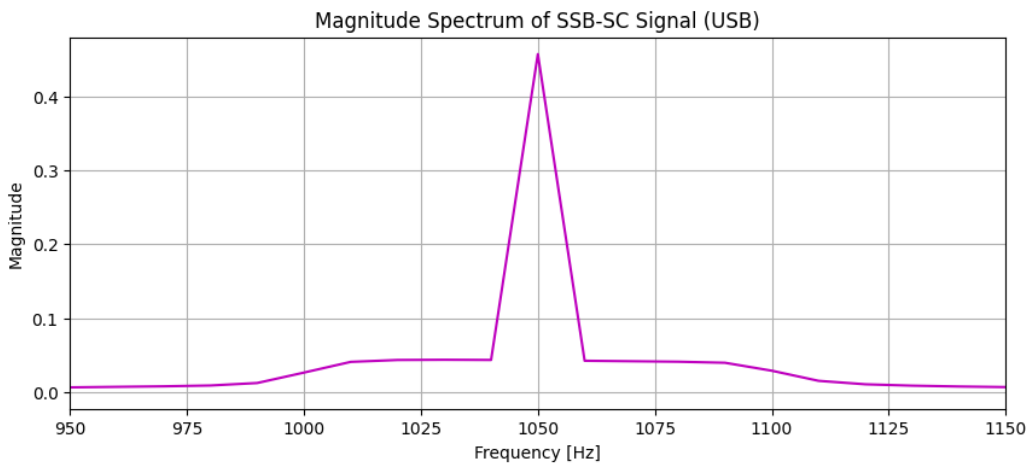

Figure 13: Plots of SSB-SC spectrum

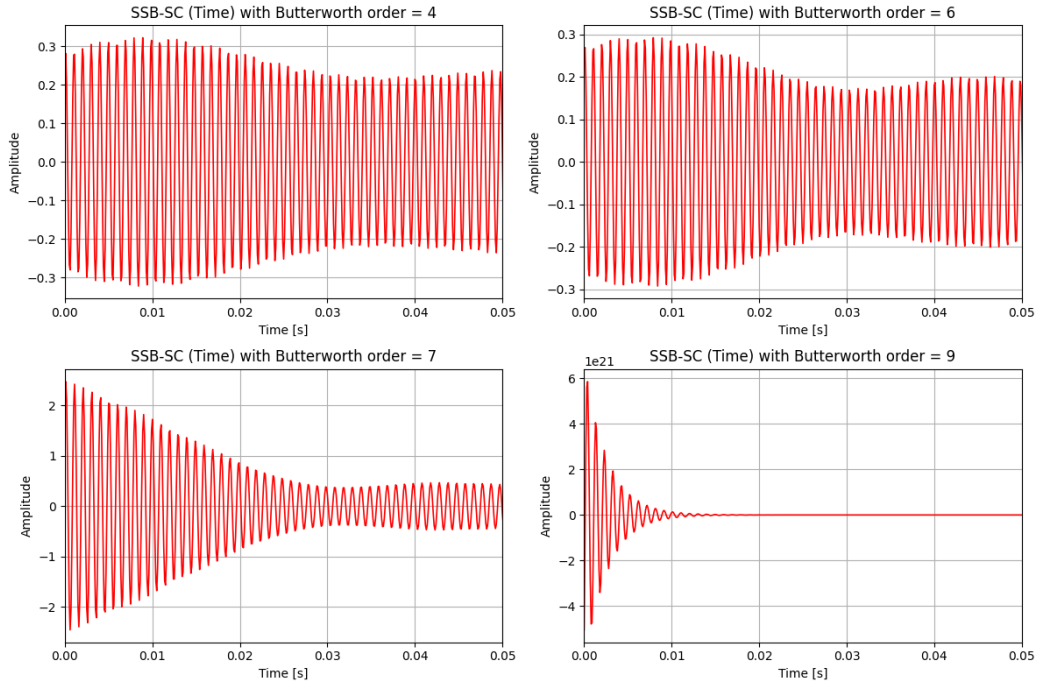

Figure 14: Time-domain waveforms of SSB-SC signal for different BW filter orders

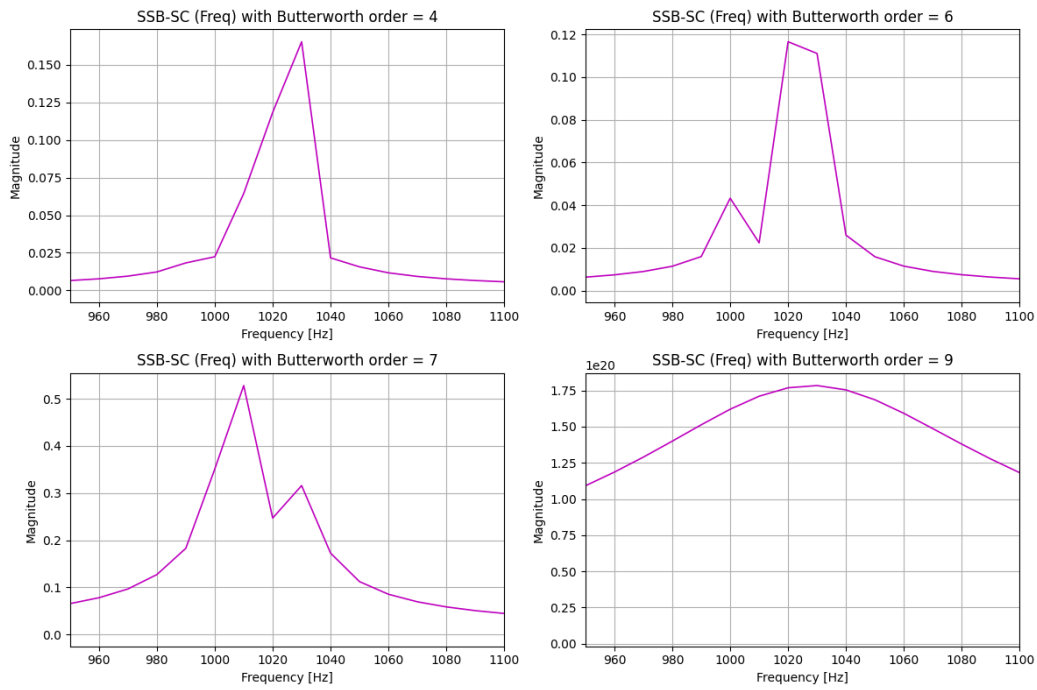

Figure 15: Frequency-domain magnitude spectra of SSB-SC signal for different BW filter orders

Observations:

The DSB-SC signal contains both upper and lower sidebands, symmetric around the carrier frequency. After applying the band-pass (BW) filter, only the upper sideband is retained,

while the lower sideband is suppressed, forming the SSB-SC signal. The time-domain plot of the SSB-SC signal shows a waveform with reduced amplitude oscillations compared to DSB-SC, as one sideband has been removed. The frequency-domain plot confirms that only the frequencies above the carrier (upper sideband) remain, while the carrier and lower sideband components are effectively suppressed. By Using this method, the bandwidth of the transmitted signal is halved compared to DSB-SC, which shows the bandwidth efficiency of SSB-SC.

Increasing the BW filter order improves upper and lower sideband separation, moving the peaks closer to the true sideband frequency. Low orders allow leakage and slightly shifted peaks, while very high orders produce cleaner sidebands with minor ripple effects.

Code (Phase Discrimination Method):

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.signal import hilbert

# Sampling Parameters
Fs = 10000                    # Sampling frequency (Hz)
T = 0.1                       # Duration of signal (s)
t = np.arange(0, T, 1/Fs)     # Time vector

# Message Signal (Baseband)
Am = 1                        # Message amplitude
fm = 50                       # Message frequency
m_t = Am * np.sin(2 * np.pi * fm * t)   # Baseband signal

# Carrier Signal
fc = 1000                      # Carrier frequency
c_cos = np.cos(2 * np.pi * fc * t)     # In-phase carrier
c_sin = np.sin(2 * np.pi * fc * t)     # (90° phase shift)

# Hilbert Transform of Message
m_hilbert = np.imag(hilbert(m_t))
```

```python

# Upper Sideband (USB)
s_usb = m_t * c_cos - m_hilbert * c_sin

# Lower Sideband (LSB)
s_lsb = m_t * c_cos + m_hilbert * c_sin

# Time-Domain Plots
plt.figure(figsize=(12, 8))

plt.subplot(3,1,1)
plt.plot(t, m_t, 'b', linewidth=1.5)
plt.title('Message Signal m(t)')
plt.xlabel('Time [s]')
plt.xlim(0, 0.05)
plt.ylabel('Amplitude')
plt.grid(True)

plt.subplot(3,1,2)
plt.plot(t, s_usb, 'r', linewidth=1.5)
plt.title('SSB-SC Signal (Upper Sideband)')
plt.xlabel('Time [s]')
plt.ylabel('Amplitude')
plt.xlim(0, 0.05)
plt.grid(True)

plt.subplot(3,1,3)
plt.plot(t, s_lsb, 'g', linewidth=1.5)
plt.title('SSB-SC Signal (Lower Sideband)')
plt.xlabel('Time [s]')
plt.xlim(0, 0.05)
plt.ylabel('Amplitude')
plt.grid(True)

plt.tight_layout()
```

```python
plt.show()

# Frequency-Domain Analysis
def plot_spectrum(signal, title, Fs):
    N = len(signal)
    S_f = np.fft.fft(signal)/N
    f = np.fft.fftfreq(N, 1/Fs)
    f_pos = f[:N//2]
    S_f_pos = 2 * np.abs(S_f[:N//2])

    plt.figure(figsize=(10,4))
    plt.plot(f_pos, S_f_pos, 'm', linewidth=1.5)
    plt.title(title)
    plt.xlabel('Frequency [Hz]')
    plt.ylabel('Magnitude')
    plt.grid(True)
    plt.xlim(fc - 100, fc + 200)
    plt.show()

# Plot spectra
plot_spectrum(s_usb, 'Magnitude Spectrum of SSB-SC (USB)', Fs)
plot_spectrum(s_lsb, 'Magnitude Spectrum of SSB-SC (LSB)', Fs)
```

Output:

The above plots are obtained using the following parameters: sampling frequency F_s = 10 kHz, signal duration T = 0.1 s, message amplitude A_m = 1, carrier amplitude A_c = 1, message frequency f_m = 50 Hz, and carrier frequency f_c = 1000 Hz.

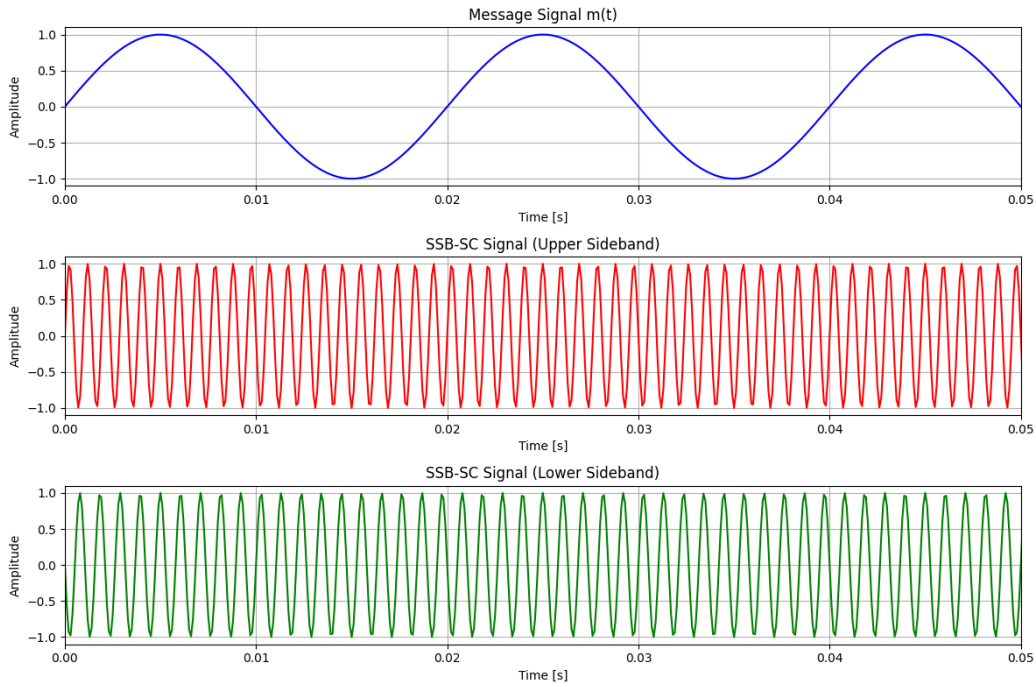

Figure 16: Plots of (a) Message signal, (b) Carrier signal, and (c) SSB-SC modulated signal.

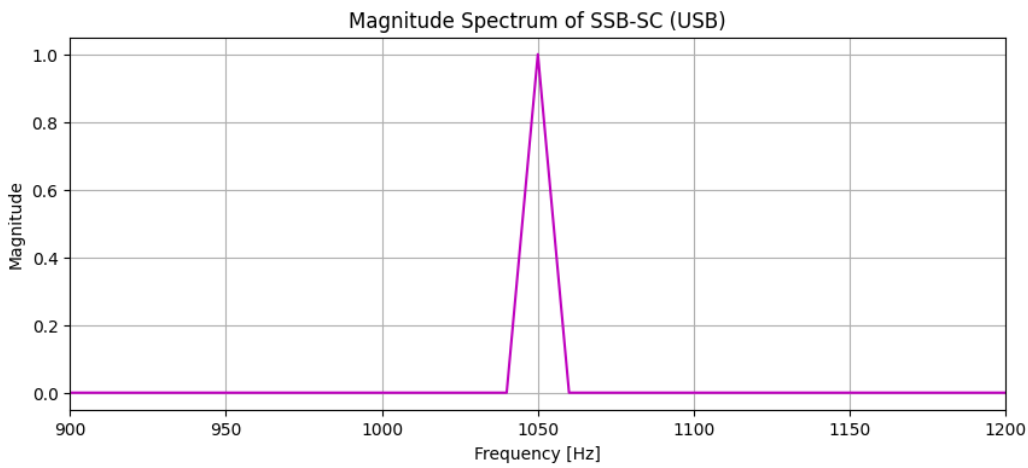

Figure 17: Frequency-domain plot of the SSB-SC signal (USB)

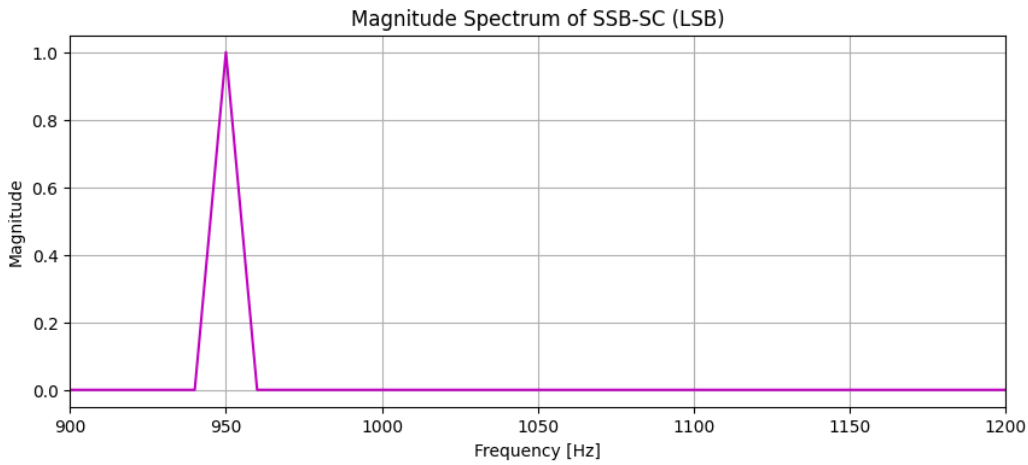

Figure 18: Frequency-domain plot of the SSB-SC signal (LSB)

Observations:

In the Phase Discrimination method for SSB-SC signal generation, the carrier is completely suppressed and only the desired sideband (either upper or lower) is retained. Unlike the Bandpass Filter (FDM) method, the sideband selection here does not depend on filter slope or order, resulting in a cleaner and more precise isolation of the single sideband. In the frequency domain, the spectrum shows a single peak corresponding to the selected sideband, demonstrating efficient bandwidth usage.

Compared to the FDM method, which can suffer from incomplete suppression of the unwanted sideband when low-order filters are used, the phase discrimination method consistently provides a clean output.

DISCUSSION

In this lab, we explored different types of amplitude modulation: DSB-FC, DSB-SC, and SSB-SC. Initially about the Modulation we learned in lab class by instructor and implemented DSB-FC then other methods were implemented as part of lab assignment.

For DSB-FC, we saw that the carrier amplitude changes according to the message signal. The carrier itself doesn't carry information but makes it easy to recover the message using simple envelope detection. We noticed that increasing the modulation index or message amplitude makes the envelope taller, and the bandwidth depends on the message frequency. For DSB-SC, the carrier is suppressed, so only the sidebands carry information. The time-domain signal looked similar to DSB-FC, but the carrier wasn't really present in the frequency spectrum. Using a longer signal duration improved the FFT resolution, making the upper and lower sidebands clearer. For SSB-SC, we learned how to remove one of the sidebands to save bandwidth. We tried both the Butterworth band-pass filter method and the phase discrimination method. The filter method showed that the filter order affects how clean the sideband is the higher the order, the steeper the filter, which gives a cleaner separation. The phase method produced a cleaner SSB signal without worrying about filter order.

CONCLUSION

The objectives of the lab were successfully achieved. From this lab, we learned Analog Amplitude modulation is all about encoding information in the amplitude of a carrier signal. Time-domain and frequency-domain plots help visualize how modulation works and how parameters affect the signals. This lab gave us a hands-on understanding of AM concepts and how engineers control power, bandwidth, and signal quality in real communication systems.